JUNE 2025

REGIONAL MARITIME UNIVERSITY

FACULTY OF ENGINEERING AND APPLIED SCIENCES

BME27, BMT27, BEE27, BCS27 & BCE27 (LEVEL 200)

BSM 202-MATHEMATICS IV

END OF SECOND SEMESTER EXAMINATION

INSTRUCTIONS: ANSWER ALL QUESTIONS FROM SECTION A. ANSWER ONE QUESTION FROM SECTION B. IN SECTION A, QUESTIONS 1 TO 10 CARRY 2 MARKS EACH, WHILE QUESTIONS 11 TO 30 CARRY 3 MARKS EACH.

TIME ALLOWED: 3 HOURS

SECTION A

1. The following equation $2xp^2 - yq = 9$ is an example of a
   A. Linear Partial Differential Equation
   B. Semi linear Partial Differential Equation
   C. Quasi linear Partial Differential Equation
   D. Nonlinear Partial Differential Equation

2. If $f(z)$ is an isogonal mapping, then
   A. It preserves the sense of angle but not magnitude
   B. It preserves the magnitude and sense of angles
   C. It preserves neither sense nor magnitude of angles
   D. It preserves the magnitude but not sense of angles

3. A critical point of the function $f(z) = z^2 + (2/z)$ is
   A. $0.5(-1 - \sqrt{3}i)$
   B. -1
   C. ± 2
   D. $\sqrt{2}$

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4. The Cauchy-Reimann equations are

A. $v_y = -u_x$ and $v_x = u_y$

B. $v_y = u_x$ and $v_x = -u_y$

C. $v_x = u_y$ and $v_x = -u_y$

D. $u_x = u_y$ and $v_x = -v_y$

5. The function $w = (az + b)/(rz + s)$ is a mobius transformation if

A. $as - br \neq 0$

B. $ar + bs = 0$

C. $as + br = 0$

D. $ar - bs \neq 0$

6. All of the following are examples of a multivalued function except

A. $f(z) = z^3$

B. $f(z) = \log z$

C. $f(z) = \tan^{-1} z$

D. $f(z) = z^{1/2}$

7. The function $f(z) = 1/(9 - z^2)$ is

A. Not Regular at $z = \pm 3i$

B. Not Regular at $z = -3$

C. Analytic at $z = \pm 3$ only

D. Holomorphic for all z

8. If $f(z)$ is an analytic function of z and if $f'(z)$ is continuous at each point within and on a closed contour, then

A. $\oint_c f(z)dz$ is non-zero

B. $\oint_c f(z)dz$ is zero

C. $\oint_c f(z)dz$ may or not be zero

D. $\oint_c f(z)dz$ is negative

9. Find the value of $\Gamma(8)/\Gamma(9)$

A. 0.125

B. 0.25

C. 0.5

D. 1

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10. The singularity of the function $f(z) = \sin(1/z)$ at $z = 0$ is
   A. Non-essential singularity
   B. Non-removable singularity
   C. A pole of order 3
   D. Essential singularity

11. What is the harmonic conjugate function of $u = 2x(1-y)$.
   A. $v = 2y + x - 2$
   B. $v = x^2 - y^2 + 2y$
   C. $v = 2y + x + xy$
   D. $v = 2(x+y)$

12. What is the value of the integral? C is the circle $|z| = 3/2$

$$\int_C \frac{z^4 - 7z^2 + 13}{(z-2)^5}\, dz$$

   A. 0
   B. 1
   C. $-\pi$
   D. 2π

13. The residue at $z = 3$ of the function

$$f(z) = \frac{7z^2}{(z+1)(z-2)(z-3)}$$

is
   A. -7
   B. 12.25
   C. 15.75
   D. 10.5

14. Given that the bilinear transformation $w = (az+b)/(cz+d)$ has only one invariant finite point, then
   A. $(a+d)^2 + 4bc = 0$
   B. $(a-d)^2 - 4bc = 0$
   C. $(a-d)^2 + 4bc = 0$
   D. $(a+d)^2 - 4bc = 0$

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15. Evaluate the following integral along the path $y = x$.

$$\int_0^{1+i} (x + iy^2)\, dz$$

A. $(6 + 5i)/5$
B. $(5 - i)/6$
C. $(5 + i)/6$
D. $(1 + 5i)/6$

16. Investigate the image of the line $y - x = -1$ under the transformation $w = 1/z$.

A. $u^2 + v^2 + u + v = 0$
B. $u^2 - v^2 - u + v = 0$
C. $u^2 + v^2 - u - v = 0$
D. $u^2 + v^2 - u + v = 0$

17. Which of the following bilinear transformations maps points $z_1 = 0, z_2 = -1$ and $z_3 = i$ of the z-plane into points $w_1 = i$, $w_2 = 0$ and $w_3 = \infty$ respectively of the w-plane?

A. $w = (3z + 2i)/(iz + 1)$
B. $w = (z + 1)/(z - i)$
C. $w = (2z + i)/(iz - 5)$
D. $w = (z + 7i)/(iz + 3)$

18. What is the equation of the image of the line $y = 4 - x$ under the transformation $w = z^2$?

A. $v = (32 - u^2)/32$
B. $v = (64 - u^2)/32$
C. $v = (128 - u^2)/32$
D. $v = (256 - u^2)/32$

19. Evaluate the following integral. C is the circle $|z| = 0.7$

$$\int_C \frac{2z + 1}{z(z + 1)}\, dz$$

A. πi
B. $-\pi i$
C. $2\pi i$
D. $3\pi i$

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0. Given that $v = x^3\sqrt{t} + (2 - 3y^2)x$ is a harmonic function, find t.

A. -2
B. 2
C. 1
D. 4

21. Given that $f(z) = \left(\frac{z+2}{z^2+4}\right)^2$, the

A. Zeroes of $f(z)$ is 2 and $z = \pm 2i$ are poles of order 2
B. Zeroes of $f(z)$ is 2 and $z = \pm 2i$ are simple poles
C. Zeroes of $f(z)$ is -2 and $z = \pm 2i$ are simple poles
D. Zeroes of $f(z)$ is -2 and $z = \pm 2i$ are poles of order 2

22. Which of the following is the general solution of the partial differential equation

$$\frac{\partial^2 z}{\partial x dt} = xt^2$$

A. $z = (x^2t^3/6) + h(t) + g(x)$
B. $z = (xt/2) + h(x) + g(t)$
C. $z = (xt^2/3) + h(t) + g(x)$
D. $z = (x^2t^2/5) + h(x) + g(t)$

$\frac{\partial z}{\partial x} = \frac{xt^3}{3}$

$z = \frac{x^2t^3}{6}$

23. Form the partial differential equation from $z = f(x^2 + y^2)$

A. $pq = x + y$
B. $px + qy = z$
C. $py - qx = 0$
D. $p^2q^2 = z$

24. Find the image of the unit circle $|z| = 1$ under the bilinear transformation

$$w = \frac{-z+2}{z-i}$$

A. $4 = u + 2v$
B. $2 = -4u + v$
C. $2 = 2u - v$
D. $3 = 2u + 4v$

$z = \frac{-z+2}{z-i}$

$z^2 - iz = -z+2$

$z^2 + z - iz - 2 = 0$

$z^2 + (1-i)z - 2 = 0$

25. All of the following are true except

A. $\Gamma(n+1) = n\Gamma(n)$
B. $\Gamma(1/2) = \sqrt{(1/2)}$
C. $\beta(m,n) = \beta(n,m)$
D. $\beta(m,n) = \frac{\Gamma(m)\cdot\Gamma(n)}{\Gamma(m+n)}$

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26. Evaluate the integral

$$\int_0^\infty x^{(1/2)} e^{-\sqrt[3]{z}}\, dx$$

A. $455\sqrt{\pi}/18$
B. $315\sqrt{\pi}/16$
C. $275\sqrt{\pi}/12$
D. $195\sqrt{\pi}/14$

$x^{n-1} e^{-\frac{3}{2}x}\, dx$

$n-1 = \frac{1}{2}$ $\quad n = \frac{3}{2}$

27. Find the characteristic curve of the Lagrange's subsidiary equation for

$$(\tan x)p + (\tan y)q = u$$

A. $(\sin x)(\operatorname{cosec} y) = k$
B. $(\sin x)(\cos y) = k$
C. $(\sec x)(\operatorname{cosec} y) = k$
D. $(\cos x)(\operatorname{cosec} y) = k$

$-x^{\frac{1}{3}}$

28. What is the solution of the PDE?

$$9\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$$

A. $u = \cos(x + 3t)$
B. $u = t^2 + 9x^2$
C. $u = \sin(3x - t)$
D. $u = e^x \cos(3x + t)$

29. The general solution of $5u_x + 3u_y = 2u$, $u(0, y) = 9e^{-5y}$.
A. $u(x, t) = 9e^{(17x/5)} \cdot e^{-5y}$
B. $u(x, t) = 9e^{(8x/3)} \cdot e^{-2y}$
C. $u(x, t) = 9e^{(13x/3)} \cdot e^{-5y}$
D. $u(x, t) = 9e^{(15x/4)} \cdot e^{-5y}$

30. The real part of the function $f(z) = z/(1 + z)$ is

A. $[x(x + y)]/[(x + 1) + y^2]$
B. $[x(y^2 + 1)]/[(x + y)^2 + y^2]$
C. $[x(x + 1) + y^2]/[(x + 1)^2 + y^2]$
D. $[x^2 + y^2]/[(x + 1)^2 + x^2]$

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SECTION B

1. Evaluate the integral

$$\int_C \frac{(z^2 - 2z)}{(z+1)^2(z^2+4)}\,dz\,,$$

C is the circle $|z| \doteq 6$.

[20]

2. Reduce into canonical form and hence find the general solution of

$$u_y + y(u_x) = x$$

[20]

3. Given that $w = \alpha + i\beta$ is the complex potential for an electric field and $\alpha = e^{2x}(x\cos 2y - y\sin 2y)$ is harmonic. Find w.

[20]

$\frac{du}{dx} =$

M.A.

$\int y\sin 2y$

$y\sin 2y$

$uv - \int v\,du$ $\quad u = y \quad v = \sin 2y$ $\quad u = y \quad v = \sin 2y$

$du = 1 \quad dv = \frac{-\cos 2y}{2}\,dy$ $\quad du = 1 \quad dv = 2\cos 2y$

$2y\cos 2y + \sin 2y$

$u = y \quad dv = \sin 2y\,dy$

$du = 1\,dy \quad v = \frac{-\cos 2y}{2}$

$\int \cos 2y$

$\frac{\sin 2y}{2}$

$-y\frac{\cos 2y}{2} - \int \frac{-\cos 2y}{2}\,dy$

$-y\frac{\cos 2y}{2} + \frac{1}{2}\int \cos 2y\,dy \Rightarrow \frac{-y\cos 2y}{2} + \frac{1}{2}\left[\frac{1}{2}(\sin 2y)\right]$